



SENIOR CAPSTONE PROPOSAL

OmniGuard SOC

*Autonomous Multi-Modal Threat Defense, Jitter-Invariant C2 Analytics
& Graph-Based Incident Triage*

PRIMARY DOMAINS

AI/ML, SecOps & Cloud

TARGET INFRASTRUCTURE

IBM LinuxONE, AWS & On-Prem

TARGET COMPLETION

December 2026 (12–14 Weeks)

THE TEAM

Team Members



Khoi

Capstone Team Member

OmniGuard SOC



Rahim

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Abu

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Roles & Responsibilities (1/2)



Khoi

Encrypted C2 Behavioral Engine & Telemetry Pipeline

- Build the SpectraC2 flow-metadata pipeline on IBM LinuxONE (s390x): IAT variance, packet symmetry, spectral density
- Train and tune the FFT/PSD feature extractor and LSTM classifier that isolate jittered C2 beacons
- Stand up the Kafka/Redis ingestion layer and Zeek/Suricata telemetry parsers feeding the graph engine



Rahim

Visual Ingress Defense & Adversary Emulation

- Develop the VerifyEye engine: YOLOv8/pHash brand-logo detection and Playwright DOM crawler
- Build the Chrome MV3 extension for real-time credential form locking on unverified submissions
- Script Flipper Zero BadUSB / rogue captive-portal scenarios and configure the Sliver C2 listener

Roles & Responsibilities (2/2)



Abu

Graph Correlation & Air-Gapped LLM Triage

- Design the Neo4j property-graph schema and entity resolution for hosts, users, IPs and processes
- Implement shortest-path, blast-radius and centrality algorithms in Cypher / NetworkX
- Deploy the air-gapped Ollama (Llama-3-Cyber) triage layer and MITRE ATT&CK mapping prompts



Aayush

Unified Dashboard & Platform Integration

- Build the React / React Flow / D3.js SOC dashboard: attack graphs, FFT heatmaps, timeline playback
- Wire WebSocket streaming for sub-second alert delivery across all three engines
- Own AWS ECS/Lambda deployment and end-to-end integration across the five project phases

MOTIVATION

Why We Chose This Topic

Enterprise SOC's defend a fractured kill chain — OmniGuard SOC unites ingress, egress and triage into one platform.



Full-Spectrum Kill Chain Defense

Over 80% of breaches start as credential harvesting, then pivot straight into C2 and lateral movement — siloed tools never connect the phases.



Neutralizing the Zero-Hour Phishing Window

Credential-theft pages live only 4–8 hours, but DNS/URL blacklists take 12–48 hours to catch up. Inline visual + DOM checks close that gap.



Overcoming the Encrypted Blind Spot

With 95%+ of traffic on TLS 1.3, DPI is obsolete. FFT periodicity and IAT metrics expose jittered beaconing without decrypting anything.



Cutting MTTD/MTTR & Preserving Sovereignty

APT dwell time averages 16+ days across fragmented logs. A Neo4j graph plus an air-gapped local LLM triages in seconds, on-prem.

Technology Stack (1/2)



Physical Ingress & Adversary Emulation

Flipper Zero (DuckyScript / Sub-GHz), Sliver C2, Cobalt Strike

Simulates BadUSB injection and rogue captive-portal redirects; delivers in-memory reverse-shell beacons to external listeners.



Visual AI & DOM Web Scanner

PyTorch, YOLOv8, ONNX Runtime, OpenCV, pHash, Playwright, Chrome MV3

Distributed crawler farm with sub-500ms brand-logo detection, DOM form validation, and inline input locking.



Encrypted C2 Behavioral Engine

IBM LinuxONE (s390x), NumPy/SciPy (FFT/PSD), Scikit-Learn, PyTorch (LSTM), FastAPI

Extracts IAT variance, packet symmetry and spectral density on s390x to isolate jittered C2 beacons.



Telemetry Streaming & Ingestion

Apache Kafka, Redis Queue, Zeek (Bro), Suricata (EVE.json), Scapy, Docker

High-throughput ingestion of connection logs and alert feeds with backpressure buffering.

Technology Stack (2/2)



Property Graph & Attack Path Discovery

Neo4j (Cypher), NetworkX, Python Graph Engine

Maintains entity relationships and runs shortest-path, blast-radius and centrality scoring.



Air-Gapped LLM Triage & Automation

Ollama, Llama-3-Cyber / Mistral, LangChain, Structured JSON Schema

Zero-cloud-egress synthesis: maps graph paths to MITRE ATT&CK and drafts remediation directives.



Unified SOC Analyst Dashboard

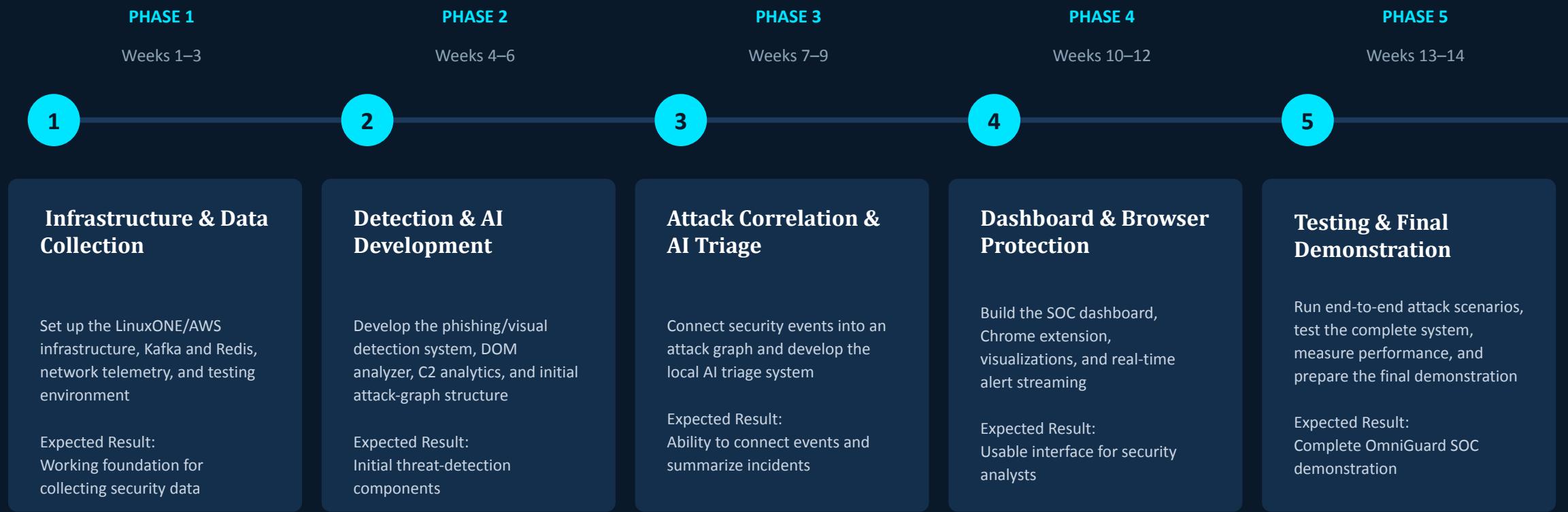
React.js, React Flow, D3.js, Tailwind CSS, Recharts, WebSockets

Interactive UI: live attack graphs, FFT heatmaps, visual brand diffs, timeline playback.

SCHEDULE

Project Timeline & Key Milestones

5-phase agile schedule — 12–14 weeks



References



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- DARPA Transparent Computing Program — Graph-based provenance tracking & APT detection via fine-grained system telemetry, 2020.
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- Oest, A., et al. (USENIX Security) — Sunrise to Sunset: end-to-end life cycle & interventions of phishing attacks.
- IEEE S&P Workshops — Cyber threat intelligence extraction using domain-adapted local LLMs without privacy leakage, 2024.
- IBM Corp. — IBM LinuxONE III Technical Guide: high-performance enterprise Linux on s390x, IBM Redbooks, 2024.
- OISF & Zeek Project — Suricata EVE JSON and Zeek network telemetry protocol parsing standards, 2024–25.